Introduction to Quarto

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1. Introduction

Quarto is a way to integrate text (markdown), computer code, figures, tables and mathematical formulas in one document.
Quarto documents (with file ending .qmd) can easily be turned into pdf format, html, website, blogs and presentations.

2. Settings for the document - the YAML

At the top of a quarto document is the so called YAML, which tells the Quarto engine how the document should be formatted. This is the YAML for this document:

```
---
title: "Introduction to Quarto"
author: "Mattias Villani"
editor: visual
---
```

which basically only gives the title and author information, everything else is using default settings. This will print (**render**) the document to html by default. If I instead wanted to render to pdf I would use the YAML:

```
---
title: "Introduction to Quarto"
author: "Mattias Villani"
editor: visual
format: pdf
---
```

3. Writing formatted text using markdown

The text written in a Quarto document can be formatted using the **markdown** language. Markdown is a very simple language for formatting text. For example, I can easily make the text **bold** by writing `**bold**` or *italic* using a single * on each side of the text. A bulleted list:

- bullet
- list
- are also
- easy

is produced by putting a dash `-` and space before each bullet point. Numbered lists are also simple, just place numbers and a dot (.) before each list item. Like this:

1. This is the first point
2. and this is second
3. and so on.

Images can be inserted by typing `![sceptic larry](/misc/larry.png){width="100"}` which would give the picture below. So the text caption for the image is written between the brackets [] and the search path to the image on your computer is given inside the parentheses ().

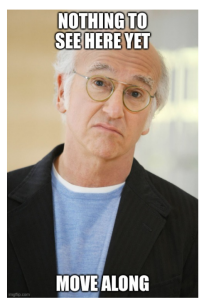

Figure 1: sceptic larry

You can leave out the figure text and the width argument and just write: `![](/misc/larry.png)`
.

Tables are also pretty easy. The table below is made by writing

```
| Name   | Age | Position            |
|-------|-----|---------------------|
| Mike   | 49  | Professor           |
```

```
| Sarah | 37  | Assistant professor |
| Anne  | 26  | PhD student         |
```

Table 1: A little table caption.

	Name	Age	Position
Mike	49	Professor	
Sarah	37	Assistant professor	
Anne	26	PhD student	

What if you want the table caption text to be below the tables, not below it? No sweat, just put `table-cap-location: bottom` in your YAML at the top of the document.

4. Writing code

A Quarto document can have executable code in them. It supports several programming languages, most notably: R, Python and Julia. You just need to add a **code chunk**. A code chunk starts and ends with three so called backticks ‘. The backtick is typed by `Shift` and the key to the left of `Backspace` on a Swedish keyboard. This is a chunk of R code:

```
```{r}
x = rnorm(100)
plot(x, type = "l", xlab = "time", main = "time series plot")
```
```

Note the little green Play-button. By pressing that we can run the code chunk. After running the code in the Quarto-document, the variables (in this case a vector `x`) will be available in the console, just like for any R code. Here is the end result of that code chunk:

```r
x = rnorm(100)
plot(x, type = "l", xlab = "time", main = "time series plot")
```